Dynamic Audio Motion Understanding: A Benchmark Suite for Structured and Causal Inference

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Abstract

Understanding the motion of sound emitting objects from audio remains an underexplored problem in machine learning, with existing work largely focused on static event classification. We introduce a synthetic benchmark suite for dynamic audio motion understanding, spanning tasks from speed and trajectory estimation to multi-source interaction and temporal reasoning. Beyond these foundational tasks, we propose advanced benchmarks including Acoustic Manifold Matching, Counterfactual Acoustic Reasoning, and Inverse Trajectory Reconstruction, which evaluate structured, causal, and continuous inference over motion. Together, these benchmarks provide a unified framework for studying motion-induced acoustic structure and enable systematic evaluation of models that aim to recover physically meaningful dynamics from audio. We demonstrate baseline results and highlight key challenges, establishing a foundation for future research in acoustic motion intelligence.

1 Initial Benchmark Suite for Audio Motion Understanding

To establish a broad and physically grounded evaluation framework for dynamic audio motion understanding, we propose an initial suite of ten synthetic benchmarks. These tasks are designed to isolate fundamental motion attributes, temporal structure, and scene level interactions that arise when sound-emitting objects move relative to an acoustic sensor. Taken together, they form a benchmark suite that measures not only whether a model can detect motion, but whether it can infer how an object moves, how multiple objects interact, and how motion evolves over time.

1.1 Benchmark 1: Speed Estimation

The first benchmark evaluates whether a model can estimate the speed of a moving sound source from an audio recording. This task may be formulated either as a regression problem, where the model predicts continuous source velocity, or as a classification problem using discretized speed bins. Speed estimation is one of the most fundamental motion inference tasks, as it is directly tied to Doppler shift, temporal envelope evolution, and changes in emitted energy over time. A successful model must learn to distinguish speed-related acoustic structure from nuisance variation such as source loudness or background noise.

1.2 Benchmark 2: Direction-of-Travel Classification

The second benchmark measures whether a model can infer the direction of motion of a sound source relative to the sensor. Example classes may include approaching, receding, left to right pass-by, right to left pass-by, or circular motion around the sensor. This task is important because it tests whether the model understands the sign and geometry of motion rather than simply detecting change in the

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waveform. In physically realistic settings, direction of travel must be inferred from asymmetric time frequency structure, amplitude progression, and spatialized motion cues when available.

1.3 Benchmark 3: Distance-of-Closest-Approach Estimation

The third benchmark focuses on estimating the minimum distance between a moving sound source and the sensor during the course of a recording. This quantity captures an essential geometric property of the motion trajectory and strongly affects the acoustic amplitude profile, spectral coloration, and local signal prominence. Unlike static distance estimation, this task requires the model to reason over the entire motion event and identify the point of nearest passage. It therefore serves as a useful test of physically meaningful geometric inference from time varying audio.

1.4 Benchmark 4: Trajectory Shape Classification

The fourth benchmark evaluates whether a model can classify the overall shape of the source trajectory. Example trajectory classes may include straight line pass by, curved pass-by, circular orbit, zig zag path, stop and go motion, or accelerating departure. This task goes beyond local motion cues and asks the model to identify the global structure of movement from the audio signature it induces. It is especially useful for distinguishing models that truly encode motion patterns from those that rely only on simple local spectral shifts.

1.5 Benchmark 5: Time-to-Event Prediction

The fifth benchmark introduces predictive temporal reasoning. Given a partial audio observation, the model must estimate the remaining time until a key future event occurs, such as closest approach, scene exit, or trajectory crossing. This task is particularly interesting because it moves beyond retrospective analysis and asks whether a model can forecast motion development from partial evidence. Such capability is important for applications in monitoring, tracking, and early warning systems where anticipating future motion matters as much as recognizing past motion.

1.6 Benchmark 6: Motion State Segmentation

The sixth benchmark reformulates motion understanding as a sequence labeling problem. Instead of assigning a single label to an entire clip, the model produces framewise motion-state labels such as approaching, nearest point, receding, stationary, occluded, or reappearing. This benchmark evaluates temporal precision and tests whether the model can localize transitions between distinct phases of motion. It also creates a bridge between audio motion analysis and structured prediction tasks commonly studied in speech and event detection.

1.7 Benchmark 7: Acceleration and Deceleration Estimation

The seventh benchmark extends motion analysis beyond constant-velocity settings by asking the model to infer whether a source is accelerating, decelerating, or moving at approximately constant speed. The task can be posed either as regression over acceleration magnitude or as classification over motion states such as braking, speeding up, or uniform motion. This benchmark is important because many real-world moving sources do not maintain constant velocity, and physically meaningful motion understanding should include higher-order kinematic structure.

1.8 Benchmark 8: Multi-Object Motion Disentanglement

The eighth benchmark considers scenes with two or more simultaneously moving sound sources. The model may be asked to estimate the number of active moving objects, infer the speed or direction of each source, or identify which source passes closest to the sensor. This benchmark significantly increases difficulty because overlapping emissions can mask one another and create ambiguous composite motion signatures. It is therefore a key test of whether a model can disentangle multiple dynamic acoustic processes within the same scene.

Benchmark	Task Type	Target Output	Evaluation Metric
Speed Estimation	Regression / Classification	Continuous velocity or discrete speed bins	MSE, MAE, Accuracy
Direction-of-Travel Classification	Classification	Motion class (approaching, receding, lateral, circular)	Accuracy, F1-score
Distance-of-Closest-Approach	Regression	Minimum source-sensor distance	MAE, RMSE
Trajectory Shape Classification	Classification	Trajectory family (straight, curved, circular, etc.)	Accuracy, F1-score
Time-to-Event Prediction	Regression	Time until closest approach or scene exit	MAE, RMSE
Motion State Segmentation	Sequence Labeling	Framewise motion states (approach, recede, etc.)	Frame Accuracy, Segment F1
Acceleration / Deceleration Estimation	Regression / Classification	Acceleration magnitude or motion state	MSE, Accuracy
Multi-Object Motion Disentanglement	Multi-task / Structured Prediction	Number of sources, per-source motion attributes	Accuracy, MAE, Assignment Error
Crossing / Interaction Event Classification	Classification	Interaction type (crossing, overtaking, etc.)	Accuracy, F1-score
Source Identity Under Motion Variation	Classification	Source or emitter class invariant to motion	Accuracy, Top-k Accuracy

Table 1: Summary of the proposed initial benchmark suite for audio motion understanding. The benchmarks span regression, classification, sequence labeling, and structured prediction tasks, covering kinematic inference, geometric reasoning, temporal prediction, and multi-source interaction.

1.9 Benchmark 9: Crossing and Interaction Event Classification

The ninth benchmark evaluates whether a model can recognize higher-level motion interactions between multiple sound sources. Example events include path crossing, overtaking, coordinated convoy like motion, temporary masking, or near simultaneous pass by. Rather than focusing on isolated source attributes, this task measures whether the model can reason about relationships among moving objects in an acoustic scene. It is especially valuable for pushing the benchmark suite beyond basic parameter estimation toward scene-level motion understanding.

1.10 Benchmark 10: Source Identity Under Motion Variation

The tenth benchmark asks whether a model can recognize source identity or source class despite large variation in motion conditions. For example, a system may be asked to identify a vehicle type, drone class, or emitter category while speed, trajectory, and closest distance vary substantially across examples. This task evaluates whether a model can disentangle intrinsic source characteristics from motion-induced acoustic distortion. It also provides a useful complement to the other benchmarks by testing whether motion-aware representations preserve source information rather than collapsing it.

1.11 Summary

These ten initial benchmarks provide a strong starting point for a synthetic evaluation suite in audio motion understanding. They cover basic kinematic inference, geometric reasoning, temporal prediction, sequence labeling, multi-source interaction, and invariance to motion-induced variation. Collectively, they define a benchmark space broad enough to support meaningful comparison across models while remaining grounded in physically interpretable acoustic phenomena. This initial suite can then be extended with more advanced benchmarks targeting robustness, uncertainty, domain transfer, inverse trajectory reconstruction, and cross-sensor generalization.

2 Advanced Benchmark: Acoustic Manifold Matching

While the initial benchmark suite focuses on estimating individual motion attributes and scene properties, more advanced tasks can evaluate whether a model has learned a structured representation of acoustic motion that generalizes across environments. To this end, we introduce the *Acoustic*

Manifold Matching benchmark, which reframes audio motion understanding as a problem of matching observations to candidate trajectory manifolds.

2.1 Problem Formulation

We consider a set of candidate motion trajectories embedded within a spatial environment, each defining a distinct mapping from latent physical state to observable audio. Let $\mathcal{M} = \{M_1, M_2, \ldots, M_K\}$ denote a collection of candidate trajectory manifolds, where each manifold M_k corresponds to a parameterized motion path (e.g., a curve in 2D or 3D space with associated kinematic properties).

Given an observed audio signal $y(t)$ generated by an unknown trajectory $M^* \in \mathcal{M}$, the task is to identify the most likely manifold:

$$\hat{M} = \arg\max_{M_k \in \mathcal{M}} P(M_k \mid y(t)). \tag{1}$$

Each manifold implicitly defines a mapping from latent state variables (position, velocity, acceleration) to acoustic observations via a forward acoustic model:

$$y(t) = H(x_{M_k}(t)) + \epsilon(t), \tag{2}$$

where $x_{M_k}(t)$ denotes the trajectory-induced state evolution and $H(\cdot)$ represents the acoustic observation operator.

2.2 Task Description

In practice, each benchmark instance consists of:

- A short audio clip generated from a moving sound source following an unknown trajectory.
- A set of K candidate trajectory segments or motion maps, each representing a plausible physical path in the environment.

The model must select which candidate trajectory best explains the observed audio. This can be interpreted as a structured classification problem over manifolds, or equivalently, as a retrieval problem in a space of motion-conditioned acoustic signatures.

2.3 Variants of the Task

To increase difficulty and probe different aspects of learned representations, several variants of the Acoustic Manifold Matching task can be defined:

Geometric Variants:

- *Simple Paths:* Straight-line or constant-curvature trajectories.
- *Complex Paths:* Piecewise or highly nonlinear trajectories with varying curvature.

Kinematic Variants:

- *Constant Velocity:* All candidates share uniform speed.
- *Variable Velocity:* Candidates differ in acceleration profiles.

Environmental Variants:

- *Anechoic:* Idealized environments with minimal reverberation.
- *Reverberant:* Realistic environments with multi-path propagation.

Ambiguity Variants:

- *Low Ambiguity:* Candidates are acoustically distinct.
- *High Ambiguity:* Candidates produce similar acoustic signatures, requiring fine-grained discrimination.

2.4 Evaluation Metrics

Performance can be evaluated using standard classification and retrieval metrics, including:

- Top-1 Accuracy
- Top-k Accuracy
- Mean Reciprocal Rank (MRR)

Additionally, structured error analysis can be performed by measuring the geometric distance between the predicted and ground-truth trajectories:

$$d(M_i, M_j) = \int \|x_{M_i}(t) - x_{M_j}(t)\| dt, \tag{3}$$

which provides insight into whether errors are physically meaningful (e.g., confusing nearby trajectories) or arbitrary.

2.5 Significance

The Acoustic Manifold Matching benchmark moves beyond estimating individual motion parameters and instead evaluates whether a model has learned a coherent representation of motion-induced acoustic structure. By requiring models to discriminate between candidate trajectory manifolds, this task directly probes the ability to invert the acoustic observation process and reason over physically plausible motion hypotheses.

This formulation also connects naturally to broader problems in inverse modeling, structured prediction, and representation learning, and provides a foundation for future extensions such as continuous trajectory retrieval, probabilistic manifold inference, and cross-environment generalization.

3 Advanced Benchmark: Counterfactual Acoustic Reasoning

Beyond estimating motion attributes or matching observations to candidate trajectories, a key question is whether a model has learned a *causal and physically grounded representation* of acoustic motion. To evaluate this capability, we introduce the *Counterfactual Acoustic Reasoning* benchmark, which probes whether a model can reason about how an audio signal would change under controlled modifications to the underlying physical process.

3.1 Problem Formulation

We consider the acoustic observation model:

$$y(t) = H(x(t)) + \epsilon(t), \tag{4}$$

where $x(t)$ represents the latent physical state (e.g., position, velocity, trajectory), $H(\cdot)$ is the acoustic observation operator, and $\epsilon(t)$ denotes noise and unmodeled effects.

In the counterfactual setting, we define an intervention on the latent state:

$$x'(t) = \mathcal{I}(x(t)), \tag{5}$$

where $\mathcal{I}(\cdot)$ represents a physically meaningful transformation such as a change in speed, trajectory, or distance.

The corresponding counterfactual audio signal is then:

$$y'(t) = H(x'(t)) + \epsilon'(t). \tag{6}$$

The goal of the benchmark is to determine whether a model can correctly reason about the relationship between $y(t)$ and $y'(t)$ induced by the intervention $\mathcal{I}$.

3.2 Task Description

Each benchmark instance consists of:

- An observed audio clip $y(t)$ generated from an unknown motion trajectory.
- A description of a counterfactual intervention $\mathcal{I}$ applied to the underlying motion or environment.
- A set of K candidate audio clips $\{y'_1(t), \ldots, y'_K(t)\}$ corresponding to possible counterfactual outcomes.

The model must select the candidate audio clip that is most consistent with the specified intervention:

$$\hat{y}' = \arg\max_{y'_k} P(y'_k \mid y(t), \mathcal{I}). \tag{7}$$

This formulation can be interpreted as a structured classification or retrieval problem over counterfactual acoustic outcomes.

3.3 Types of Interventions

To ensure physical interpretability and diversity, interventions are defined over multiple aspects of the acoustic generation process:

Kinematic Interventions:

- Changes in speed (e.g., doubling or halving velocity)
- Introduction of acceleration or deceleration
- Reversal of trajectory direction

Geometric Interventions:

- Changes in closest distance to the sensor
- Lateral shifts or mirroring of trajectories
- Modification of trajectory curvature

Environmental Interventions:

- Changes in reverberation characteristics
- Variation in signal-to-noise ratio
- Sensor or microphone response changes

Source Interventions:

- Changes in source emission strength
- Variation in source identity under fixed motion

3.4 Benchmark Variants

Several variants of the task can be defined to probe different levels of reasoning:

Multiple-Choice Counterfactual: The model selects the correct counterfactual audio from a finite set of candidates. This is the primary and most stable formulation.

Attribute-Level Counterfactual: Instead of selecting audio, the model predicts how motion attributes (e.g., speed, distance, Doppler shift magnitude) would change under the intervention.

Generative Counterfactual: The model directly synthesizes the counterfactual audio signal $\hat{y}'(t)$ given the original observation and intervention. This represents the most challenging variant and requires learning an implicit forward acoustic model.

3.5 Evaluation Metrics

Evaluation depends on the task variant:

- **Multiple-Choice:** Top-1 Accuracy, Top-k Accuracy, Mean Reciprocal Rank (MRR)
- **Attribute Prediction:** Mean Absolute Error (MAE), Mean Squared Error (MSE), or classification accuracy
- **Generative:** Spectral distance metrics, perceptual similarity measures, and physics-consistency checks

In addition, counterfactual consistency can be evaluated by verifying that predicted changes align with known physical relationships (e.g., increased speed produces stronger Doppler shifts and shorter event durations).

3.6 Significance

The Counterfactual Acoustic Reasoning benchmark extends beyond standard inference by explicitly testing whether a model captures the *causal structure* of acoustic motion. Rather than merely recognizing patterns in observed audio, the model must reason about how those patterns would change under controlled interventions.

This capability is central to robust and generalizable motion understanding, as it reflects an implicit understanding of the forward acoustic process. Furthermore, the benchmark connects naturally to broader themes in causal inference, simulation-based reasoning, and physics-informed machine learning, providing a foundation for future work in counterfactual audio modeling and decision-making under uncertainty.

4 Advanced Benchmark: Inverse Trajectory Reconstruction

While prior benchmarks evaluate motion classification, parameter estimation, and structured matching, a fundamental goal of acoustic motion understanding is to recover the underlying physical trajectory of a moving sound source directly from audio. To assess this capability, we introduce the *Inverse Trajectory Reconstruction* benchmark, which frames audio motion understanding as a continuous inverse problem over spatial trajectories.

4.1 Problem Formulation

We consider the acoustic observation model:

$$y(t) = H(x(t)) + \epsilon(t), \tag{8}$$

where $x(t) \in \mathbb{R}^d$ denotes the latent trajectory of the source in d-dimensional space, $H(\cdot)$ is the acoustic observation operator, and $\epsilon(t)$ represents noise and unmodeled effects.

The objective is to estimate the full trajectory $\hat{x}(t)$ from the observed audio signal:

$$\hat{x}(t) = \arg\max_{x(t)} P(x(t) \mid y(t)). \tag{9}$$

Unlike prior benchmarks that target discrete labels or summary statistics, this task requires reconstructing a continuous, time-varying spatial path, making it a significantly more challenging and expressive formulation.

4.2 Task Description

Each benchmark instance consists of:

- An audio clip $y(t)$ generated by a moving sound source.
- The corresponding ground-truth trajectory $x(t)$ defined over the duration of the clip.

The model must produce an estimate $\hat{x}(t)$ of the source trajectory. Depending on the formulation, this trajectory may be represented as:

- A sequence of spatial coordinates sampled at discrete time steps.
- A parametric curve (e.g., spline or polynomial representation).
- A latent representation that can be decoded into spatial coordinates.

4.3 Variants of the Task

To probe different levels of difficulty and representation learning, several variants can be defined:

Dimensionality Variants:

- *2D Reconstruction:* Planar motion relative to a fixed sensor.
- *3D Reconstruction:* Full spatial trajectory estimation.

Sensor Configuration Variants:

- *Monaural:* Single-channel audio, requiring inference from temporal and spectral cues alone.
- *Multichannel:* Stereo or array-based recordings providing spatial information.

Trajectory Complexity Variants:

- *Simple Motion:* Straight-line or constant-velocity paths.
- *Complex Motion:* Nonlinear trajectories with curvature, acceleration, or discontinuities.

Environmental Variants:

- *Anechoic:* Idealized propagation conditions.
- *Reverberant:* Realistic environments with multi-path effects.

4.4 Evaluation Metrics

Reconstruction quality can be evaluated using trajectory-level and pointwise metrics:

- **Mean Trajectory Error (MTE):**
$$MTE = \frac{1}{T} \int \|x(t) - \hat{x}(t)\| dt \tag{10}$$
- **Final Displacement Error (FDE):**
$$FDE = \|x(T) - \hat{x}(T)\| \tag{11}$$
- **Dynamic Time Warping (DTW) Distance:** Measures similarity between trajectories under temporal misalignment.
- **Velocity and Acceleration Error:** Differences between true and predicted first- and second-order derivatives of the trajectory.

In addition, geometric consistency metrics can be used to evaluate whether reconstructed trajectories preserve key properties such as closest approach distance or trajectory curvature.

4.5 Ambiguity and Uncertainty

A key challenge in inverse trajectory reconstruction is that the mapping from $y(t)$ to $x(t)$ is often non-unique. Multiple trajectories may produce similar acoustic observations, particularly in monaural or noisy settings. To account for this, the benchmark can be extended to probabilistic formulations where models output distributions over trajectories:
$$P(x(t) \mid y(t)). \tag{12}$$

Evaluation in this setting may include likelihood-based metrics or coverage of plausible trajectory sets.

4.6 Significance

The Inverse Trajectory Reconstruction benchmark represents a comprehensive test of acoustic motion understanding. Success on this task requires integrating temporal, spectral, and physical cues to recover the underlying dynamics of a moving source.

This formulation connects directly to inverse problems in physics and signal processing, and provides a natural bridge to applications such as acoustic tracking, surveillance, robotics, and scene reconstruction. Moreover, it establishes a foundation for future work in probabilistic trajectory inference, multi-source tracking, and cross-environment generalization.